

FinAuth-Audit: Validity-Aware Benchmarking of Financial Agent Authorization

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ABSTRACT

Financial agents increasingly turn weak financial priors into actions, yet a low false authorization rate (FAR) can hide coverage collapse, source-authority laundering, or insufficient evidence. We introduce FinAuth-Audit, a benchmark and release protocol for testing whether comparisons among execution rules remain valid under coverage, provenance, consequence severity, review workload, utility, and point-in-time constraints. We benchmark execution rules, not predictors. The artifact combines 124,000 controlled, provenance, and mechanistic event clusters; 2,500 event-market clusters; 936 external market dates or sessions; and a prospective 300-date actual-model partition. On a one-time controlled paper test, the lowest-FAR rule (0.064) retains authority-laundering rate (ALR) 0.245, while lineage-clean alternatives move along coverage, utility, and review frontiers. A clean current role still retains 0.172 indirect laundering, consistent with a finite-sample Bayes-risk limit under partial lineage observability. A separately preregistered test evaluates 1,200 cached proposals on 200 dates. It does not reproduce a 60-date pilot inversion: controlled-to-actual directional economic-loss ranks have Spearman $\rho = 0.657$ and exact one-sided $p = 0.932$, so the inverse hypothesis is not supported. Economic loss, material harm, and authority violation also produce different operating-point conclusions, while development-only recalibration can restore coverage and increase authority violations. A powered 500-date Binance test passes only one of two co-primary arms, so its intersection-union claim fails and is retained. FinAuth-Audit is not a model leaderboard, trading-profit benchmark, investment-advice system, or deployment claim.

CCS CONCEPTS

• Computing methodologies → Machine learning; • Applied computing → Economics.

KEYWORDS

financial agents, execution authorization, benchmark validity, provenance auditing, abstention, agent safety

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1 INTRODUCTION

Financial AI systems increasingly turn forecasts, learned signals, policy proposals, and language-model assessments into *weak financial priors*: structured beliefs that may inform an action but do not

by themselves confer execution authority. Examples include a language model extracting a directional assessment from an earnings release, a forecast model proposing a position change, or a risk system flagging a credit or liquidity condition. In each case the generator proposes; a separate execution rule decides whether to execute, reduce, route to review, or abstain. Existing financial benchmarks primarily evaluate question answering, forecasting, language understanding, or task completion [6, 32–35]. Agent benchmarks extend evaluation to tool use and multi-step interaction [20, 22, 24, 37], but successful prediction or task completion does not establish that a prior deserves execution authority. We therefore benchmark execution rules and the validity of their evaluation, not predictors.

Figure 1 gives the running example. Cost-Aware authorizes 36.7% of frozen paper-test rows and has the lowest FAR, 0.064, but ALR 0.245. Hard Role Gate authorizes 75.0% with ALR zero and leads conservative validation hypervolume, but its FAR is 0.656. Lifecycle has FAR 0.147, ALR zero, coverage 0.293, and review rate 0.429. No Action receives FAR N/A, not zero. The benchmark therefore exposes a three-way frontier among outcome harm, lineage integrity, and usable coverage rather than declaring a universal winner.

Low FAR can be statistically reproducible and still be invalid as an authorization ranking because coverage, provenance, review load, and generator mechanism change the evaluation object.

Benchmark outcomes can be fragile to task selection and measurement design [8, 17, 19, 25], while high-stakes data work can accumulate undocumented assumptions and shortcuts [13, 15, 23, 26]. FinAuth-Audit addresses three coupled blind spots. First, low FAR can be obtained by suppressing difficult cases or ignoring provenance. Second, clean current-role metadata can hide upstream authority laundering. Third, external-transfer and training-utility claims can be promoted despite insufficient power, collapsed mechanism coverage, or a preregistered mixed result.

The benchmark is organized around three contribution axes. **Coverage-aware certification** asks whether raw safety rankings survive joint FAR, ALR, coverage, utility, and review-load reporting. **Provenance under partial observability** asks what current roles, full lineage, and missing lineage can identify under delegation, paraphrase, role noise, and multiple hops. **External and training validity gates** ask whether point-in-time evidence and learned authorization comparisons satisfy predeclared power, effect-size, and mechanism-coverage conditions before positive claims are permitted.

FinAuth-Audit is a benchmark asset and release protocol, not a new execution rule. It contains 124,000 controlled, provenance, and mechanistic event clusters; public event and order-book layers; and a separately frozen 300-date, five-asset, two-task actual-model partition. The latter contains 50 development dates, a one-time 200-date paper test with 1,200 cached proposals, and 50 author-unevaluated

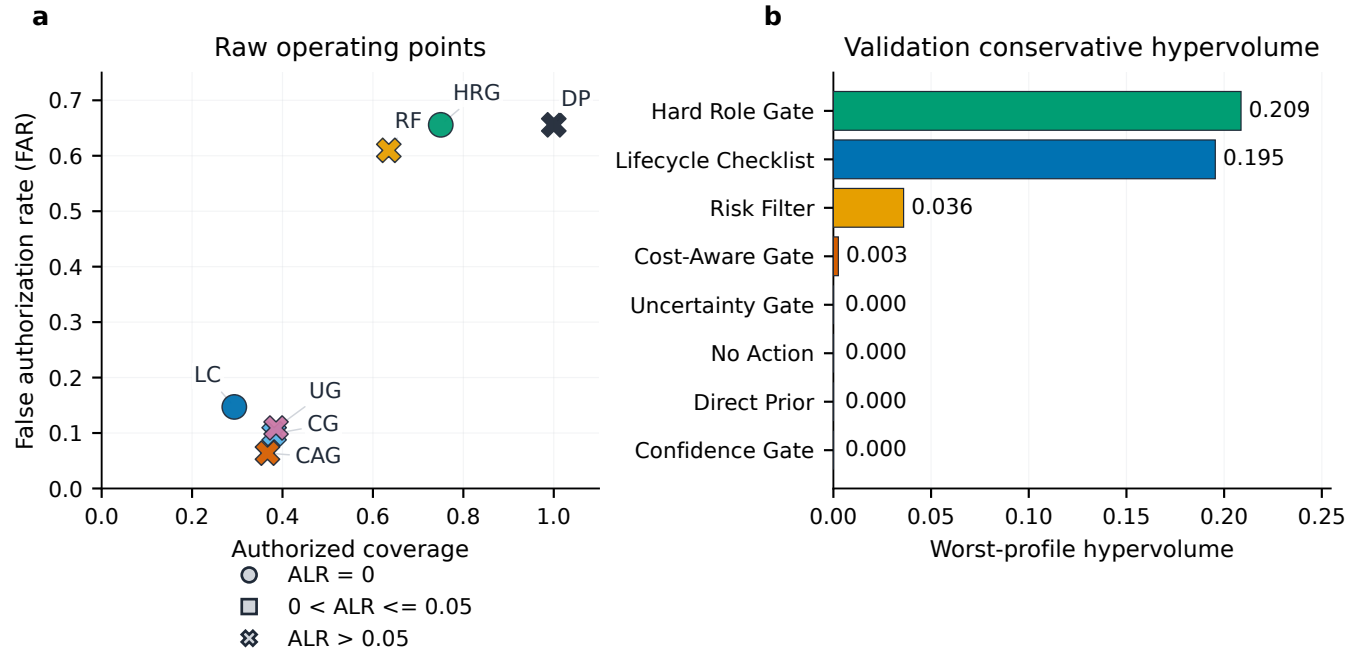


Figure 1: Raw FAR, lineage integrity, coverage, and conservative hypervolume produce different rule orderings. Panel (a) reports one-time paper-test operating points; marker shape encodes ALR strata. Short labels are DP (Direct Prior), CG (Confidence Gate), UG (Uncertainty Gate), RF (Risk Filter), CAG (Cost-Aware Gate), HRG (Hard Role Gate), and LC (Lifecycle Checklist). Panel (b) is a pre-frozen validation sensitivity, not a second paper-test ranking. Zero authorization produces FAR and ALR N/A, not zero.

holdout dates whose proposal plaintext remains encrypted. Latent opportunity, source reliability, costs, stress, and provenance are sampled before rule outputs. Before outcome inspection, existing test clusters were split deterministically into one-time paper tests and community-hidden challenges; evaluators, rules, thresholds, data hashes, and protocols were bound to content-addressed archives.

Our contributions are:

- a validity-aware benchmark for execution rules across weak financial priors, with joint coverage, FAR, ALR, utility, missed-opportunity, and review-workload reporting;
- branch-neutral utility, sequential-market, and institutional-workflow generators with cluster splits, mechanism hold-outs, and a one-time frozen paper/community protocol;
- a provenance task with a finite-sample observability limit, direct and indirect laundering, traceable and untraceable delegation, safe-delegation coverage, and structural N/A; and
- pre-result power, effect-size, and mechanism-coverage gates plus a content-addressed artifact that retains a mixed order-book failure and a larger prospective actual-model test that does not reproduce the inverse rank-transfer pattern observed in a 60-date pilot.

EPV serves as one equal-information baseline; removing it does not change any headline conclusion. FinAuth-Audit does not claim deployment readiness, trading profitability, investment advice, public transfer, or a globally optimal rule.

2 RELATED WORK

Financial and agent benchmarks. Financial benchmarks evaluate numerical reasoning, language understanding, forecasting, and decision support [6, 21, 32–34]. General agent benchmarks evaluate task completion, browsing, and tool use [20, 22, 24, 36, 37]. These resources establish important capability measurements, but their evaluation unit is generally a prediction, answer, or completed trajectory rather than the validity of an execution-authority benchmark. Supplementary Table 4 locates FinAuth-Audit along that distinction.

Selective decisions and safe evaluation. Selective classification studies prediction with abstention [11, 14]; conformal methods provide distribution-free uncertainty statements under stated assumptions [1, 27, 31]; off-policy evaluation estimates policy value from logged behavior [10, 18, 29]; and safe reinforcement learning studies constrained sequential decisions [12, 28]. FinAuth-Audit does not replace these methods. It audits whether a benchmark can distinguish genuine authorization quality from zero coverage, authority leakage, and invalid transfer.

Benchmark and model-risk validity. Benchmark fragility, behavioral testing, measurement validity, data documentation, and shortcut learning motivate stronger evaluation contracts [8, 13, 15, 17, 19, 25, 26]. Financial model-risk guidance likewise emphasizes validation, controls, and traceable reporting [2, 4]. The closest

decision-rule perspective evaluates choices across many weak experiments [3, 7, 30]. FinAuth-Audit changes the unit from business experiments to weak financial priors and adds execution coverage, source provenance, point-in-time transfer, and mechanism-level training validity.

3 TASK AND VALIDITY THREATS

3.1 Authorization object

Each row contains a weak financial prior p_i attached to an event cluster $c(i)$ and a decision-time information set I_{t_i} . Prior generators may be heuristics, forecast models, learned policies, language-model agents, or evidence-only risk systems. The prior specifies a candidate direction, confidence, expected edge, uncertainty, horizon, source identity, and source role. An execution rule g maps only legal fields in (p_i, I_{t_i}) to one of four routes:

$$d_i = g(p_i, I_{t_i}) \in \{\text{execute, reduce, review, abstain}\}.$$

Execute and reduce are authorized actions. Review is a deferred workflow and is not execution coverage. Realized return, post-cost utility, harm, tail loss, lineage truth, and future decoys are evaluation fields and are prohibited from deployable rules.

Let A_i indicate an authorized action, U_i denote selected post-cost utility, $H_i = \mathbb{1}[A_i = 1 \wedge U_i < 0]$, and L_i indicate that an authorized action originated from an ineligible source. For an evaluation profile S ,

$$\text{Cov}(S) = \frac{\sum_{i \in S} A_i}{|S|}, \quad \text{FAR}(S) = \frac{\sum_{i \in S} H_i}{\sum_{i \in S} A_i}, \quad \text{ALR}(S) = \frac{\sum_{i \in S} L_i}{\sum_{i \in S} A_i}.$$

FAR and ALR are N/A when $\sum A_i = 0$. We define review load as

$$\text{Rev}(S) = \frac{1}{|S|} \sum_{i \in S} \mathbb{1}[d_i = \text{review}].$$

Cumulative authorized utility (CAU) sums U_i over authorized rows; missed opportunity cost (MOC) sums the best positive eligible utility on non-authorized rows. Safe-delegation coverage measures authorization of eligible delegated positive opportunities. These quantities describe different objectives and are not collapsed into a scalar leaderboard.

Reader map. Coverage measures how often a rule executes or reduces. FAR measures harmful outcomes conditional on authorization; ALR measures ineligible authority lineage on the same denominator. Review load counts cases deferred to a person. CAU and MOC track authorized and missed positive utility. Structural N/A means that a denominator does not exist, not that risk is zero.

For the multi-task actual-model layer, FAR is retained only as a legacy alias for *directional economic-loss authorization*, $\Pr(U < 0 \mid A = 1)$. We separately report material harm, defined by task-specific predeclared severity thresholds; directional tail harm; and authority violation, defined as authorization of an evidence-only risk critic. A losing action is therefore not automatically a material or authority failure, and a profitable action cannot legalize an ineligible source. Structural N/A also applies when no development candidate satisfies a preregistered operating-point constraint.

Observable-provenance limit. Let $L \in \{0, 1\}$ denote laundering truth and let X be any legal observable signature available to a

provenance rule. A standard Bayes-classification identity [9] gives

$$\inf_h \Pr[h(X) \neq L] = \sum_x \min\{\Pr(X = x, L = 0), \Pr(X = x, L = 1)\}. \quad (1)$$

Under equal class priors this is $(1 - \text{TV}(P_{X|L=0}, P_{X|L=1}))/2$. Thus identical legal signatures with conflicting lineage truth impose unavoidable error on every rule measurable with respect to X ; randomization cannot improve the Bayes risk. FinAuth-Audit reports the plug-in finite-sample value and its exact overlap identity, not a claim that deployed lineage distributions are known. The proof is in the supplement.

3.2 Four validity threats

Coverage collapse occurs when a rule suppresses difficult profiles, leaving an apparently low FAR over a narrow authorized subset. **Provenance laundering** occurs when the current source appears eligible although authority originated from an ineligible source and propagated through delegation, paraphrase, or multiple hops. Full lineage, partial signatures, and current-role-only observations define different identifiability regimes. **Invalid transfer or training evidence** occurs when public replay violates point-in-time availability, treats derived rows as independent, lacks event-cluster power, or compares learners after mechanism-specific coverage collapses. FinAuth-Audit treats these as benchmark-validity failures, not merely model errors.

4 BENCHMARK CONSTRUCTION

4.1 Pipeline and data layers

Supplementary Figure 4 summarizes the artifact. Each of 50,000 utility-IID clusters contains four source-role variants, producing 200,000 rows split by cluster into 120,000 training, 40,000 validation, and 40,000 test rows. Before outcome inspection, the 10,000 test clusters were ordered by a SHA-256 rule and split exactly into 5,000 paper-test and 5,000 community-hidden clusters. The seven frozen opportunity slices include ordinary, moderate stress, severe stress, rare safe opportunity, high missed opportunity, stress-only period, and a no-opportunity control.

Branch-neutral construction. The generator follows a fixed sequence:

- (1) sample an event state, opportunity intent, stress severity, liquidity, and cost regime;
- (2) sample four source-role-specific noisy priors and confidence/uncertainty fields;
- (3) realize return, full/reduced post-cost utility, and outcome-only harm labels;
- (4) attach provenance transformations or task-specific workflow state; and
- (5) run execution rules only after the row and legal-field manifest are frozen.

An oracle-positive opportunity indicates that at least one role- and provenance-eligible full or reduced action has positive post-cost utility. It is never a deployable feature. Generator parameters, expected opportunity rates, mechanism holdouts, and seeds are frozen before evaluation; validation defects remain in the negative-results log and were repaired only by rule-independent corrections.

To test generator dependence, we prospectively add two mechanisms without changing rule thresholds: a sequential-market generator with regimes, inventory, market impact, and path dependence, and an institutional-workflow generator with trading, credit, and research-to-action tasks, risk budgets, compliance flags, approval depth, and review pressure. Together they add 96,000 rows from 24,000 clusters. They are mechanistically distinct but still author-controlled scenarios, not market simulators or institutional logs.

4.2 Provenance layer

The provenance layer reuses the controlled market-state scaffold but adds role claims, verified roles, original source identifiers, delegation parents, transformation chains, content hashes, hop depth, traceability, and lineage attestation. Five attack families are balanced at the cluster level: clean, role noise, delegation, paraphrase, and multi-hop. Role noise rates are 0%, 5%, 10%, 20%, and 40%; hop depths range from one to five. Traceable rows expose enough lineage for detection in principle. Untraceable rows intentionally omit that evidence and are reported as a separate stress/impossibility stratum.

4.3 Point-in-time public, order-book, and training layers

The public layer contains 2,500 structurally valid Polymarket event clusters, split chronologically into 1,500 training, 500 validation, and 500 sealed test clusters. The source probability is the last observation strictly before a fixed decision timestamp at 35% of scheduled lifecycle. Outcome and post-decision fields are labels only. Twenty-three high-activity months reached the frozen pagination cap and are disclosed as truncated. The pre-result power gate estimates 0.164 power, with LCB₉₅ 0.160, against a 0.80 requirement, so the layer remains exploratory_only. FRED and GDELT remain documented extensions rather than confirmatory sources.

A v0.3 extension adds observed market-outcome replay with constructed priors and roles: 185,592 Binance rows from 703 UTC dates and 16,776 licensed MES rows from 233 RTH sessions. Binance combines official banded depth with one-minute BTCUSDT/ETHUSDT bars, split 120/500/83 into development, confirmatory paper test, and hidden dates; planning power uses the frozen v0.2 variance model. MES uses one-contract BBO replay split 60/140/33, with the paper test descriptive only. Both sources reuse frozen priors, rules, and thresholds; outcomes remain sealed until one-time evaluation, and MES rows are not redistributed.

A v0.6 actual-proposal extension uses 300 non-overlapping dates from May 2025 through March 2026, deterministically assigned across BTC, ETH, BNB, SOL, and XRP. Each date contains a directional-execution task with an eligible edge proposer and a 25% risk-limit-increase task whose source role is deterministically assigned as eligible proposer or evidence-only critic before generation. Three model configurations receive the same legal pre-decision context. The chronological split contains 50 development, 200 one-time paper-test, and 50 author-unevaluated holdout dates. Holdout proposals were generated and encrypted before outcome evaluation; their key remains outside the repository

and the ciphertext was not decrypted. Outputs retain malformed placeholders without repair or favorable reruns. Rationales are audit metadata and are excluded from every rule.

The registered training layer freezes D0–D7 at 400 clusters per seed and five seeds. A primary endpoint is valid only when every held-out mechanism has a 95% coverage lower bound of at least 0.05. A prospective v0.3 robustness study preserves that primary and adds Logistic Regression, histogram gradient boosting, MLP, and selective Logistic learners; 400, 1,000, 5,000, and 20,000-cluster budgets; three curricula; and six held-out mechanisms. The validity condition is deliberately allowed to make a comparison undefined.

5 EVALUATION PROTOCOL

5.1 Rules and legal information

The controlled audit compares No Action, Direct Prior, confidence and uncertainty gates, a risk filter, a cost-aware gate, a hard current-role gate, and a lifecycle checklist. The provenance audit adds No Role Gate, Shared Threshold, Soft Penalty, Provenance Hard Gate, Provenance Learned Gate, and an equal-information EPV Adapter. Equal-information means that the adapter receives the same legal decision-time fields as comparable rules, without outcomes or privileged source evidence. It represents one learned utility-aware rule family and tests whether low FAR transfers to provenance integrity; it receives no privileged status. These baselines represent distinct authorization designs, and none defines the benchmark identity. Each implementation declares its legal fields, and the feature-access audit fails if a deployable rule uses realized return, utility, harm, laundering truth, tail loss, future decoys, or public outcomes.

5.2 Cluster-aware bounds and certification

All uncertainty resamples event clusters, not derived prior rows [5]. Main controlled and provenance results use 5,000 bootstrap replicates. For each rule and profile, FinAuth-Audit estimates a one-sided 95% coverage lower bound and FAR/ALR upper bounds. The frozen certification grid uses

$$\tau_h \in \{.05, .10, .15, .20, .25, .30\},$$

$$\tau_a \in \{0, .01, .02, .05\},$$

$$c_{\min} \in \{.05, .10, .20, .30, .50\}.$$

A cell passes only if

$$\text{UCB}_{.95}(\text{FAR}) \leq \tau_h, \quad \text{UCB}_{.95}(\text{ALR}) \leq \tau_a, \quad \text{LCB}_{.95}(\text{Cov}) \geq c_{\min}.$$

Certification volume is the passed fraction of the 120 deterministic cells, not a multiple-testing result or weighted leaderboard score. We report the minimum of overall and stress-profile volumes. Coverage collapse is $1 - \text{Cov}_{\text{stress}}/\text{Cov}_{\text{ordinary}}$ and remains undefined when ordinary coverage is zero.

Because the discrete grid encodes risk preferences, we also report five frozen profile grids, a continuous conservative hypervolume, and the FAR/ALR/coverage/MOC Pareto set on validation. These diagnostics do not select a universal winner. Review is reported separately because routing a case to a person consumes capacity: the artifact evaluates fixed 5%, 10%, 20%, and 40% review-capacity scenarios and normalized 0.5–5 basis-point review-cost sensitivities.

Table 1: FinAuth-Audit releases multiple validity layers rather than a single aggregate row count. Counts and boundaries are generated from manifests. Controlled, provenance, external order-book, and actual-model results use separately frozen one-time paper tests; the event-market and mechanistic/training extensions retain their stated validation-only or sealed status.

Layer	Rows	Clusters	Split or budget	Coverage / boundary
Controlled core	200,000	50,000	30k / 10k / 5k / 5k clusters	train / val. / paper / hidden
Provenance laundering	200,000	50,000	30k / 10k / 5k / 5k clusters	5 attacks; 2 trace strata
Mechanistic generators	96,000	24,000	sequential / institutional	validation robustness; test sealed
Public point-in-time	2,500	2,500	1.5k / 0.5k / 0.5k	500 validation reported; test sealed
External order book	202,368	936	Binance 120/500/83; MES 60/140/33	powered public + licensed descriptive
Actual model proposals	1,200	200	50 / 200 / 50 dates	3 models; 2 tasks/date; aggregate only
Training utility	144 fits	400–20k / curriculum	3 curricula; 4 learners	6 mechanisms; N/A retained

5.3 Pre-result gates

Public evaluation is permitted to become confirmatory only if point-in-time checks, independent-cluster counts, rule coverage, and the frozen power threshold all pass. The gate uses validation uncertainty and structural test-cluster counts without evaluating public test outcomes. Training utility uses a mechanism-wise FAR endpoint at a validation-fixed target coverage. The endpoint is defined only when every held-out mechanism passes its coverage-LCB floor. D2–D7 are compared with D1 using Holm correction [16]; D0 and the role-neutral control are outside that family. Undefined seed pairs remain absent rather than being imputed as wins or losses.

The v0.6 zero-shot transfer test compares six shared directional rules using economic-loss authorization. Support for the preregistered inverse hypothesis requires both date-bootstrap $\Pr(\rho < 0) \geq .95$ and an exact one-sided inverse-or-more-extreme permutation $p \leq .05$. Registered diagnostics include Kendall τ_b , leave-one-rule-out, pairwise reversals, and model, asset, and volatility strata; subgroup estimates are not powered confirmatory tests. A secondary in-domain track selects unchanged rule forms from fixed grids using five chronological folds over 50 development dates and a 0.10 execute-or-reduce coverage floor. Selections are frozen before paper outcomes; invalid selections remain N/A.

5.4 One-time paper test and hidden challenge

After all rules, thresholds, metrics, generators, robustness profiles, and reporting templates were fixed, a deterministic SHA-256 ordering split the 10,000 controlled/provenance test clusters into 5,000 paper-test and 5,000 community-hidden clusters using identifiers only. A validation rehearsal reproduced registered point metrics to within 1.1×10^{-15} . The evaluator and scientific surface were then stored in a content-addressed archive with SHA-256 be95f4d1...98aa8; a one-time registry records command, environment, inputs, outputs, and inspection time. The community-hidden, public-test, and FinAuth-Worlds hidden outcomes remain unevaluated. Strict verification checks 475 conditions, including surface hashes and the no-community-access flag.

6 BENCHMARK FINDINGS

Evidence tiers. Controlled and provenance headline results use the one-time 5,000-cluster paper test. Separately frozen tests use 500 powered Binance UTC-date clusters, 140 descriptive MES sessions,

and 200 dates with 1,200 actual-model proposals across two tasks. Event-market transfer, cross-generator, and training-robustness analyses remain prospective validation evidence; every community-hidden partition is unevaluated.

6.1 Finding 1: Raw FAR can misrank authorization rules

Table 2 reports the controlled paper test. Direct Prior authorizes every row and has FAR 0.656 and ALR 0.250. Cost-Aware Gate has the lowest raw FAR, 0.064, and CAU 40.77, but ALR 0.245 and certification volume zero. Lifecycle Checklist has higher FAR, 0.147, and lower CAU, 30.14, but ALR zero and the only nonzero minimum profile certification volume, 0.30. It obtains that operating point with coverage 0.293 and review rate 0.429. No Action has coverage zero and FAR/ALR N/A; it cannot win by abstaining everywhere.

The ranking is not an artifact of this partition: validation-to-paper-test Spearman correlation is 1.000 for raw FAR rank and effectively 1.000 for certified rank. Yet the pre-frozen use-case families and continuous sensitivity do not select the same operating point: no rule passes the conservative family, Lifecycle has nonzero balanced and coverage-first volume, and Hard Role Gate has larger conservative hypervolume than Lifecycle. Seven nonzero-action rules lie on the exact four-dimensional Pareto set, so the supplement reports the frontier rather than using Pareto membership as a discriminator. **Finding 1: FinAuth-Audit exposes a multi-objective authorization frontier; low raw FAR is neither necessary nor sufficient for a valid rule comparison.**

6.2 Finding 2: Current roles do not establish provenance

Figure 2 and Table 3 report the provenance paper test. Hard Role Gate has direct leakage zero but indirect leakage and ALR 0.172. Its current source is eligible, but the authority chain need not be clean. Provenance Hard Gate sets ALR to zero at coverage 0.591, FAR 0.674, safe-delegation coverage 0.698, and review rate 0.183. The learned gate raises coverage to 0.603 while accepting ALR 0.018. Utility-screening rules can have low FAR while retaining indirect laundering; Lifecycle has FAR 0.146 and ALR 0.165; EPV has FAR 0.039 and ALR 0.169.

Equation (1) gives the same boundary from another angle. The plug-in current-role-only Bayes risk is 0.129 on traceable rows and

Table 2: Paper-test operating points and validation conservative hypervolume reject a single-winner interpretation. CAU is cumulative authorized utility; MOC is missed opportunity cost; Val. HV is a pre-frozen validation sensitivity. Full certification grids and the Pareto set appear in the supplement. N/A denotes structural zero authorization.

Rule	Cov.	FAR	ALR	Review	CAU	MOC	Val. HV
No Action	0.000	N/A	N/A	0.000	0.00	31.20	0.000
Direct Prior	1.000	0.656	0.250	0.000	20.38	0.00	0.000
Confidence Gate	0.382	0.099	0.250	0.000	39.46	0.00	0.000
Uncertainty Gate	0.386	0.109	0.248	0.000	39.31	0.01	0.000
Risk Filter	0.635	0.610	0.209	0.000	23.31	5.97	0.036
Cost-Aware Gate	0.367	0.064	0.245	0.000	40.77	0.01	0.003
Hard Role Gate	0.750	0.656	0.000	0.000	16.01	0.00	0.209
Lifecycle Checklist	0.293	0.147	0.000	0.429	30.14	0.63	0.195

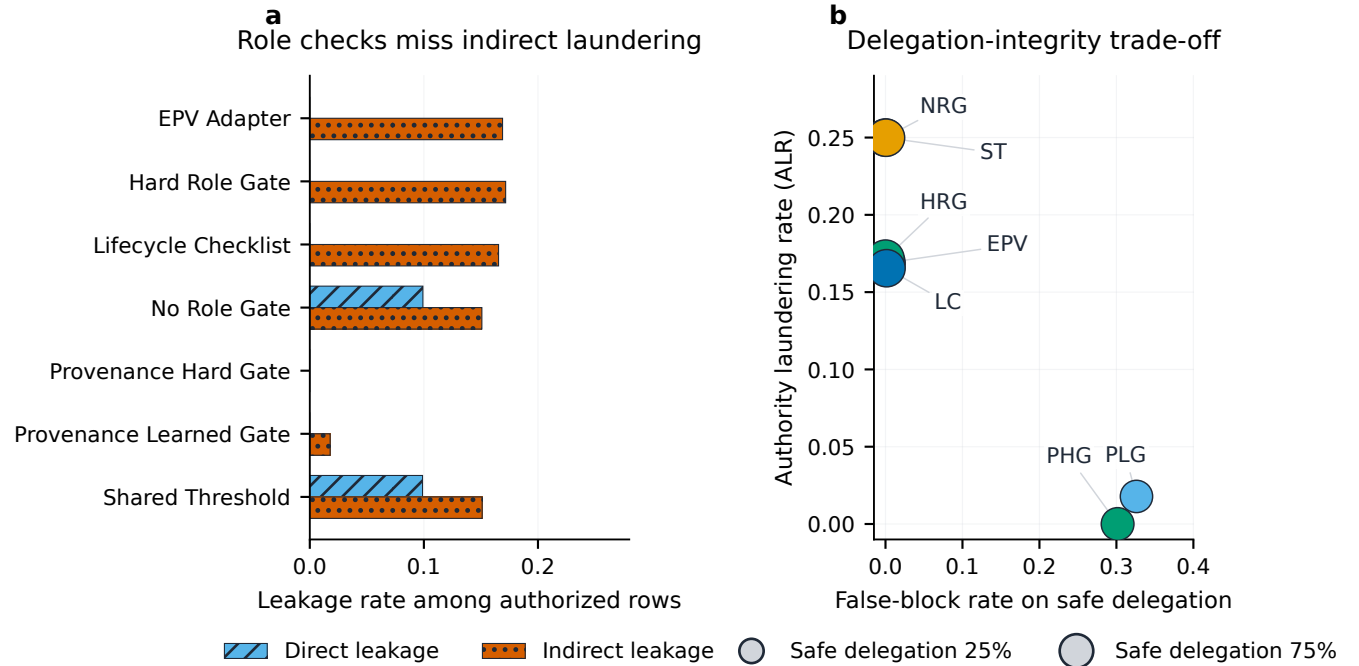


Figure 2: Current-role integrity, provenance integrity, and usable delegation differ on the paper test. Panel (a) separates direct from indirect leakage using color and hatch. Panel (b) plots ALR against false blocking; bubble area encodes safe-delegation coverage. Labels abbreviate the rules listed in Table 3.

0.256 on untraceable rows; legal partial signatures reduce it to 0.061 and 0.083. The implementation independently verifies the overlap identity to numerical precision. These values quantify ambiguity on the registered validation distribution; they do not license reconstruction of hidden institutional lineage. **Finding 2: role eligibility and provenance integrity are distinct evaluation targets, and missing lineage creates an identifiable information cost rather than a zero-risk state.**

6.3 Finding 3: A larger prospective test does not reproduce the pilot inversion

The v0.6 paper test evaluates 1,200 cached proposals from three model configurations on two tasks across 200 independent UTC

dates and five assets. The preregistered zero-shot inverse-rank hypothesis is not supported: controlled-to-actual directional economic-loss ranks have Spearman $\rho = 0.657$ and Kendall $\tau_b = 0.600$, the exact one-sided permutation test gives $p = 0.932$, and only 0.173 of 10,000 date-cluster bootstrap replicates have $\rho < 0$. A 60-date pilot had reported $\rho = -0.928$; the larger prospective result is binding and the pilot is not retained as a headline finding. Only 3 of 15 rule pairs reverse, all involving Lifecycle, and omitting Lifecycle gives $\rho = 1.000$ among the remaining five rules. Nine malformed expected records remain deterministic non-action placeholders in task denominators; none is repaired or rerun. The wide cluster-bootstrap interval measures rank instability across date resamples, while the exact test targets the preregistered inverse alternative. Model, asset,

Table 3: Paper-test provenance controls trade authorization quality, lineage integrity, delegation coverage, and review load. Direct and indirect columns are leakage rates among authorized rows.

Rule	Cov.	FAR	ALR	Direct	Indirect	Safe del.	Review
No Action	0.000	N/A	N/A	N/A	N/A	0.000	0.000
No Role Gate	1.000	0.674	0.250	0.099	0.151	1.000	0.000
Shared Threshold	0.371	0.122	0.250	0.099	0.151	0.999	0.000
Soft Penalty	0.406	0.198	0.238	0.085	0.153	0.999	0.000
Hard Role Gate	0.879	0.674	0.172	0.000	0.172	1.000	0.121
Provenance Hard Gate	0.591	0.674	0.000	0.000	0.000	0.698	0.183
Provenance Learned Gate	0.603	0.668	0.018	0.000	0.018	0.674	0.100
Lifecycle Checklist	0.335	0.146	0.165	0.000	0.165	0.999	0.296
EPV Adapter	0.297	0.039	0.169	0.000	0.169	0.999	0.000

and volatility estimates remain underpowered diagnostics rather than substitute hypotheses.

Endpoint choice changes interpretation. Zero-shot economic-loss authorization lies between 0.641 and 0.694 among authorizing rules, while material harm ranges from 0.138 to 0.238 and authority violation from zero to 0.088. Lifecycle has the lowest material and tail-harm rates and zero authority violation, but coverage 0.121, review rate 0.069, and substantial missed benign opportunity. Development-only recalibration raises Confidence coverage from 0.103 to 0.331 and Uncertainty coverage from 0.086 to 0.370, while their authority-violation rates rise from 0.032 to 0.091 and from 0.049 to 0.092. Lifecycle remains structural N/A: its best development candidate reaches coverage 0.08, below the frozen 0.10 floor. Thus threshold adjustment does not supply a universal transport repair.

The separately preregistered mechanistic generators provide validation-only robustness evidence. With unchanged thresholds, sequential-market and institutional-workflow raw FAR ranks have Spearman 0.994, whereas their correlations with utility IID are 0.683 and 0.738. The lowest-FAR rule differs from the highest-certification rule in two of three generator families, and the sequential generator certifies no rule under the frozen surface. **Finding 3: a validity-aware benchmark must be able to retire an attractive pilot claim, distinguish endpoint semantics, and report localized transfer sensitivity without replacing a failed primary with favorable subgroups.**

6.4 Finding 4: Validity gates preserve mixed and undefined results

The preregistered 500-date Binance IUT **fails** despite adequate power (0.970/0.965). Cost-Aware minus Lifecycle lowers false-authorization burden by -0.0170 (95% CI $[-0.0198, -0.0138]$), passing the registered -0.015 arm, while laundering rises only $+0.0154$ (95% CI $[+0.0140, +0.0168]$), below the pre-result $+0.020$ SESOI. Seven-day blocks agree, and a favorable 4-bps sensitivity remains secondary. MES is descriptive only: a fixed six-prior bundle makes the rule pair differ on one ineligible critic, so its $1/6$ contrast is not an estimand (Supplementary Table 11).

The earlier event-market layer still passes timestamp, field-access, split-isolation, and provenance checks, but its estimated cluster-level power is 0.164 with LCB95 0.160, below the frozen

0.80 requirement. It remains exploratory_only, and its public test stays sealed.

The registered 400-cluster Logistic study has no valid D2–D7 versus D1 comparison because at least one held-out mechanism violates the 5% coverage-LCB floor. The prospective v0.3 study increases learner capacity and budget without altering that primary. At 20,000 clusters, full-audit Logistic and selective Logistic recover five of six mechanisms on average (0.833 valid-mechanism fraction), but sequential-market coverage still collapses. The complete cross-mechanism FAR remains N/A for every learner/curriculum combination. **Finding 4: a validity-aware benchmark is useful when it retains a powered mixed failure, an underpowered transfer layer, and a coverage-collapsed learner comparison instead of converting any of them into a positive claim.**

Supplementary Figure 5 visualizes the powered mixed external result and the incomplete mechanism-coverage surface without changing their registered statuses.

7 USE, LIMITATIONS, AND RESEARCH OPPORTUNITIES

Community use. External researchers can implement a rule against the legal feature manifest and run the same cluster-aware certification, provenance, power, and mechanism-coverage checks. The release records field access, four-way routes, hashes, and structural N/A. The 5,000-cluster community-hidden partition remains unevaluated under versioned maintenance.

From controlled evaluation to deployment. The controlled layers are authorization scenarios, not price-path simulations, institutional logs, or human labels. Branch-neutral construction and the frozen test improve internal validity, while prospective generators expose mechanism sensitivity. Because all generators share authorship and legal-field design, conclusions remain conditional on the released scenarios and do not establish practitioner agreement or deployment readiness.

External and review limits. Binance adds observed spread and depth. Planning variance is transported, constructed priors remain in v0.3, and only one arm passes. The v0.6 layer adds actual model proposals across five assets and two tasks, but legal source roles remain benchmark-assigned and the study does not observe institutional permission. Its 200 dates support aggregate inference but not precise model- or asset-specific rankings. The

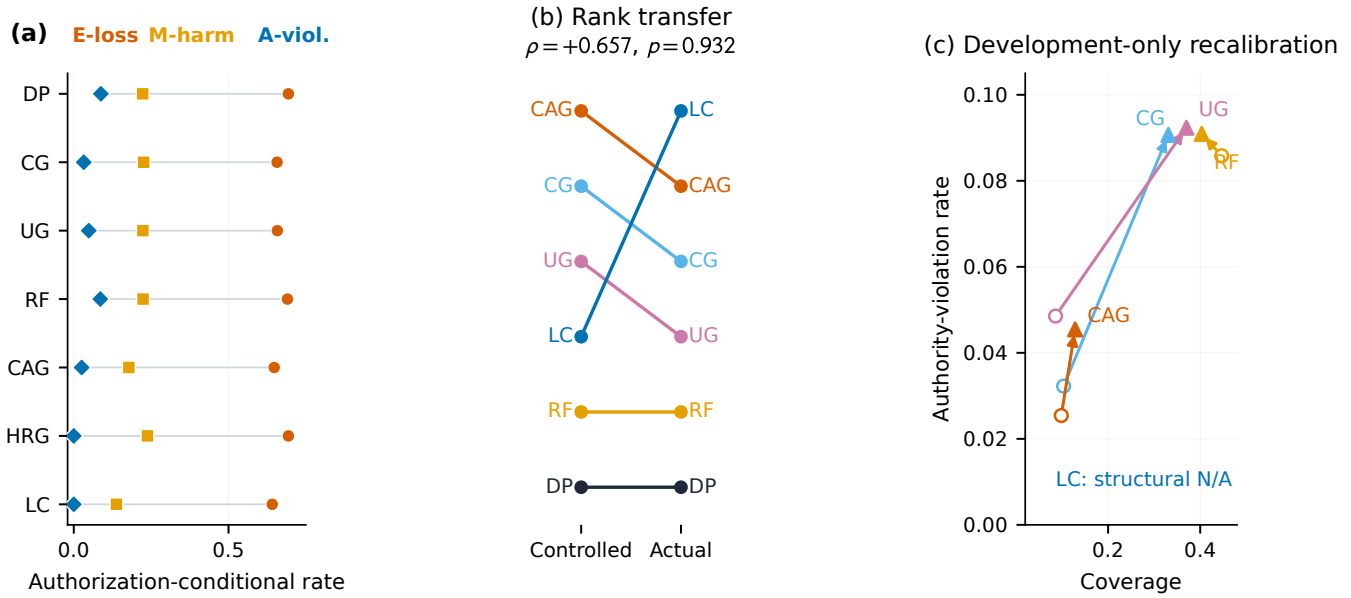


Figure 3: The actual-model paper test separates consequence semantics, rank transfer, and threshold transport. Panel (a) reports zero-shot authorization-conditional rates: economic loss is directional post-cost loss, material harm uses preregistered task-specific severity, and authority violation is authorization of an evidence-only critic. Panel (b) reports the preregistered aggregate rank test; the inverse hypothesis is not supported. Panel (c) uses open circles for zero-shot and triangles for development-only recalibration. Lifecycle is structural N/A because none of 36 candidates met the 0.10 development coverage floor. This is rule-transfer evidence, not a model or trading leaderboard.

holdout snapshot is self-custodied ciphertext rather than third-party escrow, and model aliases can drift after generation. MES is licensed, single-instrument, underpowered, and has one non-estimand role contrast. Event-market, FRED, and GDELT evidence remains non-confirmatory; no source provides practitioner labels or broad generalization. Review load and cost are simulated, not observed staffing performance.

What the validity gates show. Across 144 prospective fits, broader data recover five of six mechanisms, while sequential-market coverage remains unresolved. The actual-proposal audit shows why the release protocol matters: a dramatic 60-date pilot inversion did not survive the larger preregistered test, and development-only threshold recalibration did not uniformly improve authority quality. Coverage-constrained objectives, independently authored generators, practitioner-labelled authorization tasks, and stronger point-in-time data can be added without changing the frozen paper test. FinAuth-Audit remains an offline research benchmark, not investment advice or institutional approval.

8 CONCLUSION

FinAuth-Audit makes benchmark validity part of financial-agent authorization evaluation. Its frozen controlled test shows that low raw FAR can coexist with authority laundering, while provenance integrity, delegation coverage, utility, missed opportunity, and review load form a genuine frontier. A larger prospective actual-proposal test does not reproduce the inverse rank transfer observed in a 60-date pilot; it instead separates economic loss, material harm, authority violation, and development-only recalibration. The powered order-book test retains a mixed intersection-union failure, while public-power and training-coverage gates keep weaker evidence exploratory or N/A. The public artifact preserves each evaluation surface and its adverse results. FinAuth-Audit therefore supports reproducible comparison of execution rules across weak financial priors without treating prediction quality, any model source, any baseline, or any single metric as execution authority.

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A ARTIFACT AND REPRODUCTION

The aggregate v0.6 release combines the frozen v0.2 controlled/provenance surface, prospective robustness extensions, the separately frozen v0.3 external order-book surface, and the frozen v0.6 actual-model proposal extension. Every data and result layer records a configuration hash, data hash, split registry, seed registry, output hashes, and claim boundary. The aggregate verifier transitively checks the earlier surfaces and the v0.6 freeze-to-result chain; the repository suite is rerun before release. The controlled paper test is bound to content-addressed archive be95f4d1...98aa8 and took 80.59 seconds on the development machine with peak resident memory about 550 MB. Re-running any inspected one-time paper test is prohibited except for a versioned correctness defect that preserves the original output.

The figure generator writes PDF, SVG, and PNG from one source and records a minimum visible font size of 8 pt. Generation fails if marked point labels overlap. The table generator reads frozen result files and records input/output hashes in `table_manifest.json`; no reported numeric result is hand-entered into the manuscript. A page-faithful HTML bundle mirrors the combined PDF, while the benchmark card, reproduction guide, maintenance policy, and external-submission protocol state the community-use contract.

B CONTROLLED CERTIFICATION DETAILS

The controlled evaluation excludes the explicit no-opportunity slice from certification but reports it as a negative control. Table 7 gives the cluster-bootstrap bounds used by the frozen 120-cell surface. A zero-action bootstrap replicate contributes coverage zero and execution-denominator metrics *N/A*. If every replicate is undefined, no dependent cell passes.

The paper-test opportunity rates are ordinary 0.450, moderate stress 0.386, severe stress 0.236, rare safe opportunity 0.131, high missed opportunity 0.507, stress-only period 0.195, and no-opportunity control 0.000. The oracle-positive label is outcome-only and is never available to a rule.

C PROVENANCE DEFINITIONS AND STRATA

Direct leakage authorizes a currently ineligible source. Indirect leakage authorizes a currently eligible source whose original authority is ineligible. ALR uses the same authorized-action denominator as indirect laundering truth. Safe-delegation coverage is the fraction of eligible delegated positive opportunities that remain authorized; false block is its complement. Table 8 separates the traceable and untraceable strata.

Traceable rows include the attested lineage-role chain needed to detect laundering in principle. Untraceable rows omit that evidence. On the paper test, Provenance Hard Gate has traceable coverage 0.724 and untraceable coverage zero; its untraceable FAR and ALR are undefined rather than zero. The empirical current-role-only Bayes error lower bound rises from 0.129 on traceable rows to 0.256 on untraceable rows, while legal partial signatures reduce the corresponding bounds to 0.061 and 0.083. This is an evidence-availability boundary, not a claim that a detector should infer hidden lineage.

Proof of Equation (1). For a fixed observable signature x , any deterministic rule h predicts either zero or one. Its error mass at x is therefore either $\Pr(X = x, L = 1)$ or $\Pr(X = x, L = 0)$, and the smaller achievable mass is their minimum. Summing independently over signatures proves the first equality. A randomized rule takes a convex combination of the same two error masses and cannot improve the minimum. Under equal priors, write $p_\ell(x) = \Pr(X = x \mid L = \ell)$; then

$$\begin{aligned} \sum_x \min\{\tfrac{1}{2}p_0(x), \tfrac{1}{2}p_1(x)\} &= \tfrac{1}{2} \left(1 - \tfrac{1}{2} \sum_x |p_0(x) - p_1(x)| \right) \\ &= \tfrac{1}{2} (1 - \text{TV}(P_0, P_1)). \end{aligned}$$

The audit uses empirical signature frequencies in place of joint probabilities. Its minority count is the exact minimum empirical error for rules using that signature, not a distribution-free bound for another institution or period.

D PUBLIC POINT-IN-TIME GATE

The Polymarket metadata window is 2022-01-01 through 2026-07-17. Monthly sampling caps each month at 120 clusters and each series/month at 12 clusters. Twenty-three high-activity months reached the frozen pagination limit and are disclosed as truncated. Historical probability comes from CLOB price history when available and documented public trades otherwise. The fixed decision time is 35% of scheduled lifecycle; the source probability precedes the decision, and the first execution observation follows it but precedes resolution.

The first CLOB-only design produced sufficient histories for 516 candidate markets. A documented read-only trades fallback increased structural usability without creating synthetic probabilities, spread, or depth. An earlier 3,000-cluster design covered only about 19 hours and was rejected as externally narrow before result inspection. Both negative design iterations remain in the project log.

E EXTERNAL ORDER-BOOK REPLICATION

The separately versioned v0.3 extension freezes public Binance USD-M futures banded depth plus one-minute bars and licensed Databento MES BBO before outcome evaluation. It is observed market-outcome replay with constructed priors and assigned roles. Binance uses 500 UTC-date paper-test clusters; planning power under the frozen v0.2 variance model is 0.970 and 0.965 for the two registered SESOIs. MES uses 140 RTH-session clusters and is classified as descriptive because its corresponding design powers are 0.528 and 0.514. Feature rows and sealed execution inputs are separate; returns, post-cost utility, harm, and burden are materialized only after freeze verification inside the write-once evaluator.

Table 4: FinAuth-Audit treats benchmark validity as the evaluation object. “Rules” denotes explicit comparison of execution or decision rules; the remaining columns denote validity audits, not generic safety features.

Benchmark	Evaluation unit	Rules	Coverage audit	Provenance	Power gate
FinQA [6]	financial QA	no	no	no	no
PIXIU [34]	financial tasks	no	no	no	no
FinBen [33]	financial tasks	no	no	no	no
AgentBench [20]	agent tasks	no	no	no	no
WebArena [37]	web tasks	no	no	no	no
Weak experiments [7]	experiment portfolios	yes	no	no	no
FinAuth-Audit	weak financial priors	yes	yes	yes	yes

Table 5: Artifact components and machine-checkable release paths.

Artifact	Release path	Verification
Paper test	results/paper_test/	one-time registry; 5k clusters
Community hidden	sealed/community_hidden/	5k clusters; unevaluated
Freeze archive	manifests/paper_test_snapshots/	content-addressed SHA-256
Point-in-time layer	results/public_audit/	power and timestamp gates
External order book	results/external_orderbook_v03/	powered mixed result; 550 checks
Actual-model paper test	results/real_agent_v06/paper_test/	3 models; 200-date, 2-task aggregate test
Historical-memory audit	results/real_agent_v06/historical_memory_audit/	outcome-blind aggregates; no rationale text persisted
Training robustness	results/training_robustness_v03/	144 fits; 6 mechanisms
Figures	paper/figures/	PDF, SVG, PNG; text QA
HTML and cards	paper/html/; docs/	combined paper/supplement; local assets
Release archive	dist/finauth-audit-0.6.0.tar.gz	allowlist; clean-room hash and split scan
Verifier	verify_artifact.py	all frozen surfaces, fields, hashes, and tests

Table 6: Pre-frozen policy families and continuous hypervolume do not define a universal winner. Values are worst-profile validation pass fractions over each author-specified grid. The conservative family certifies no rule; Lifecycle has nonzero balanced and coverage-first volume, while Hard Role Gate has the largest continuous conservative hypervolume. These are sensitivity diagnostics, not practitioner-elicited thresholds.

Rule	Conservative	Balanced	Coverage-first	Continuous HV
No Action	0.000	0.000	0.000	0.000
Direct Prior	0.000	0.000	0.000	0.000
Confidence Gate	0.000	0.000	0.000	0.000
Uncertainty Gate	0.000	0.000	0.000	0.000
Risk Filter	0.000	0.000	0.000	0.036
Cost-Aware Gate	0.000	0.000	0.000	0.003
Hard Role Gate	0.000	0.000	0.000	0.209
Lifecycle Checklist	0.000	0.375	0.188	0.195

FinAuth-Audit evaluates whether authorization benchmarks themselves are valid

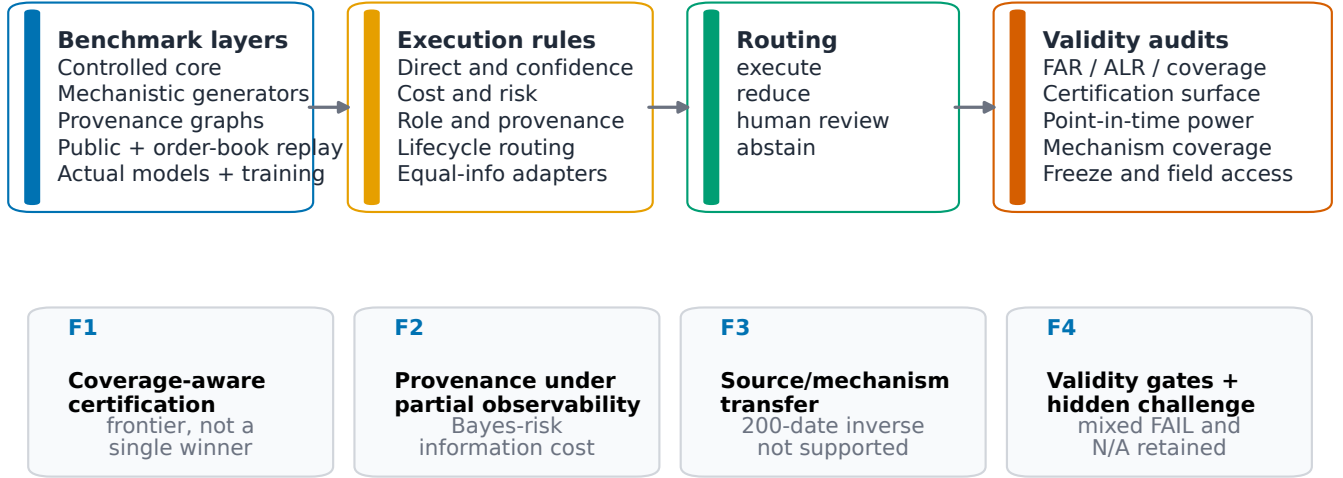


Figure 4: FinAuth-Audit separates benchmark layers, execution rules, routing outcomes, and validity gates. Controlled rows support certification; provenance rows perturb authority chains; public rows are restricted by decision-time availability; and matched training corpora test mechanism coverage. The v0.6 F3 box is generated from the frozen primary-support flag and records that inverse rank transfer is not supported.

Table 7: One-sided cluster-bootstrap bounds and certification volumes by rule and profile on the paper test.

Rule	Profile	Coverage LCB	FAR UCB	ALR UCB	Cert. volume
No Action	overall	0.000	N/A	N/A	0.000
No Action	stress	0.000	N/A	N/A	0.000
Direct Prior	overall	1.000	0.667	0.250	0.000
Direct Prior	stress	1.000	0.712	0.250	0.000
Confidence Gate	overall	0.370	0.107	0.253	0.000
Confidence Gate	stress	0.341	0.161	0.254	0.000
Uncertainty Gate	overall	0.375	0.117	0.250	0.000
Uncertainty Gate	stress	0.346	0.174	0.250	0.000
Risk Filter	overall	0.625	0.624	0.211	0.000
Risk Filter	stress	0.469	0.678	0.177	0.000
Cost-Aware Gate	overall	0.356	0.069	0.247	0.000
Cost-Aware Gate	stress	0.302	0.053	0.248	0.000
Hard Role Gate	overall	0.750	0.667	0.000	0.000
Hard Role Gate	stress	0.750	0.713	0.000	0.000
Lifecycle Checklist	overall	0.285	0.156	0.000	0.300
Lifecycle Checklist	stress	0.233	0.127	0.000	0.400

Table 8: Traceability-specific provenance outcomes on the paper test. N/A is structural when a rule authorizes zero rows in a stratum.

Rule	Traceability	Cov.	ALR	Safe del.	False block
EPV Adapter	traceable	0.288	0.148	0.999	0.001
EPV Adapter	untraceable	0.338	0.248	0.998	0.002
Hard Role Gate	traceable	0.851	0.150	1.000	0.000
Hard Role Gate	untraceable	1.000	0.254	1.000	0.000
Lifecycle Checklist	traceable	0.325	0.145	0.999	0.001
Lifecycle Checklist	untraceable	0.378	0.244	0.998	0.002
No Action	traceable	0.000	N/A	0.000	1.000
No Action	untraceable	0.000	N/A	0.000	1.000
No Role Gate	traceable	1.000	0.249	1.000	0.000
No Role Gate	untraceable	1.000	0.254	1.000	0.000
Provenance Hard Gate	traceable	0.724	0.000	1.000	0.000
Provenance Hard Gate	untraceable	0.000	N/A	0.000	1.000
Provenance Learned Gate	traceable	0.627	0.015	0.676	0.324
Provenance Learned Gate	untraceable	0.497	0.035	0.668	0.332
Shared Threshold	traceable	0.370	0.249	1.000	0.000
Shared Threshold	untraceable	0.376	0.252	0.999	0.001
Soft Penalty	traceable	0.404	0.234	1.000	0.000
Soft Penalty	untraceable	0.416	0.254	0.999	0.001

Table 9: Legal observability controls the minimum empirical laundering-classification error. Joint separation is the weighted empirical separation term in Equation (1); the implementation verifies the corresponding overlap identity to numerical precision.

Traceability	Observable signature	Rows	Collision frac.	Bayes risk	Joint sep.
traceable	current role only	32,857	1.000	0.129	0.742
traceable	legal partial signature	32,857	0.720	0.061	0.877
untraceable	current role only	7,143	1.000	0.256	0.488
untraceable	legal partial signature	7,143	0.519	0.083	0.834

Table 10: Public-source classifications and the pre-result power gate. The point-in-time layer remains exploratory_only because power is below the frozen requirement; no public test outcome is reported.

Layer / source	Clusters	Power	LCB95	Req.	Classification
Public point-in-time	500 (structural)	0.164	0.160	0.80	exploratory_only
polymarket					descriptive public-derived
polymarket_repeated					excluded from inference
gdelt					controlled information-shock
fred					exploratory non-vintage
polymarket_point_in_time					confirmatory candidate; structural only

Table 11: The external intersection-union claim fails, and observed market data do not make weak priors or authority roles observational. Panel A reports Cost-Aware Gate minus Lifecycle Checklist. Binance is confirmatory; MES is licensed descriptive evidence, and its fixed six-prior laundering contrast is structural. Panel B separates observed, pre-decision derived, benchmark-assigned, and sealed outcome-only fields.

Panel A: frozen external replication outcomes.

Source	Evidence	Clusters	Endpoint	Diff.	95% CI	SESOI	Status
Binance	powered public	500	False-auth. burden	-0.017	[-0.020, -0.014]	≤ -0.015	pass
Binance	powered public	500	Laundering burden	+0.015	[+0.014, +0.017]	$\geq +0.020$	fail
Binance	powered public	500	Intersection-union	–	–	both arms	FAIL (1/2)
MES	licensed descriptive	140	False-auth. burden	-0.019	[-0.029, -0.010]	≤ -0.015	descriptive
MES	licensed descriptive	140	Laundering burden	1/6 by design	–	not applied	not an estimand

Panel B: field origins and evaluation boundary.

Origin	Fields	Evaluation boundary
Observed source	quotes, bars, depth, timestamps	available at or before the registered decision time
Pre-decision derived	momentum, volatility, imbalance, spread/cost proxies	deterministic functions of observed source fields
Benchmark assigned	prior family, action, confidence, source role, eligibility	frozen weak-prior construction; not observed agent behavior
Outcome only	entry/exit prices, return, utility, harm, burden	sealed until the one-time evaluator; illegal for deployable rules

Sensitivity and release boundary. The Binance seven-day moving-block sensitivity preserves the arm-level interpretation. At the frozen primary 8-bps fee, only false-authorization burden passes. A 4-bps sensitivity passes both arms, whereas 12 bps passes neither; these fee-grid results use fewer bootstrap replicates and remain secondary. No sensitivity replaces the registered primary. Binance releases source archives, official checksums, derived rows, manifests, and code. MES reproduction requires Databento entitlement, and row-level licensed data are not redistributed. MES is an optional licensed replication layer; all headline benchmark functionality and primary controlled/provenance evaluations are reproducible without redistributing MES records. The public dataset entry, exact schema, date/session definitions, reconstruction scripts, manifests, and checksums are documented in the artifact.

F ACTUAL MODEL PROPOSAL EXTENSION

The v0.6 extension evaluates actual cached proposal outputs rather than deterministic agent stubs. GPT-5.4-mini, GPT-5.5, and Claude Sonnet 4.6 receive the same legal pre-decision context on 300 independent dates assigned across BTC, ETH, BNB, SOL, and XRP. Each date includes a directional-execution task and a 25% risk-limit-increase task. Directional sources are eligible edge proposers; risk-limit source role is assigned by frozen event-ID parity as an eligible proposer or evidence-only critic before generation. The

chronological partition is 50 development, 200 one-time paper test, and 50 author-unevaluated holdout dates. The holdout contexts were sent to the same model configurations before outcome evaluation; 300 proposal rows were encrypted, their key remains outside the repository, and the ciphertext was not decrypted.

The development/paper cache contains 1,500 plaintext proposal rows and has raw schema validity 0.994. Nine expected records remain malformed placeholders, there are zero repairs, and no outcome-conditioned or favorable rerun is permitted. One extra record was ignored under the preapproved duplicate/extra record policy. Rationales and self-reported evidence are retained in private audit caches but prohibited from every execution rule and excluded from the aggregate public release. The explicit historical-memory lexical audit finds no registered retrospective phrase, but cannot establish absence of implicit model memory.

The preregistered zero-shot inverse rank-transfer criterion is not supported; the independent unit is one UTC date ($n = 200$). The point estimate is $\rho = 0.657$ with Kendall $\tau_b = 0.600$; the bootstrap median and 95% interval are 0.543 [-0.657, 0.943], and $\Pr(\rho < 0) = 0.173$. Only 3 of 15 pairwise orderings reverse, all involving Lifecycle and near-tied differences. Model-specific point estimates range from -0.029 to 0.714 and asset-specific estimates from -0.143 to 0.812, with intervals spanning zero; these are underpowered diagnostics, not model or asset rankings. The earlier 60-date v0.5 pilot value -0.928 is retained only as the motivation for the prospective design and is superseded for the inverse-transfer claim.

The recalibrated track is explicitly secondary. Confidence and Uncertainty recover coverage but increase authority violation; Risk Filter changes little; Cost-Aware gains modest coverage; Lifecycle remains undefined. These results separate zero-shot threshold transport from in-domain threshold selection, but do not establish a universally transferable rule form. The extension does not support model superiority, institutional permission, deployment readiness, investment advice, or trading profitability.

Table 12: Economic loss, material harm, and authority violation are distinct zero-shot endpoints. The table reports overall operating points across 200 UTC-date clusters. Economic loss applies to directional authorization; material harm pools preregistered task-specific severity; authority violation enforces the declared evidence-only critic boundary. Utility is normalized task utility, not profit.

Rule	Dates	Cov.	Econ. loss	Material harm	Tail harm	Authority viol.	Utility	Review
Direct Prior	200	0.457	0.694	0.223	0.078	0.088	-0.155	0.000
Confidence Gate	200	0.103	0.657	0.226	0.088	0.032	-0.029	0.000
Uncertainty Gate	200	0.086	0.658	0.223	0.053	0.049	-0.021	0.000
Risk Filter	200	0.447	0.690	0.224	0.079	0.086	-0.154	0.000
Cost-Aware Gate	200	0.098	0.648	0.178	0.019	0.025	-0.023	0.000
Hard Role Gate	200	0.417	0.694	0.238	0.078	0.000	-0.144	0.000
Lifecycle Checklist	200	0.121	0.641	0.138	0.015	0.000	-0.025	0.069

Table 13: Task-specific zero-shot and development-recalibrated operating points retain structural N/A. Bounds use 10,000 UTC-date cluster-bootstrap replicates over 200 dates. Model abstention remains in the task denominator but cannot be authorized. Lifecycle has no recalibrated point because no development candidate meets the frozen coverage floor.

Track	Task	Rule	Cov. [95%]	Task harm [95%]	Authority viol.	Utility
Zero-shot	Directional	Direct Prior	0.642 [0.590, 0.692]	0.694 [0.620, 0.765]	0.000	-0.262
Zero-shot	Directional	Confidence Gate	0.170 [0.135, 0.205]	0.657 [0.537, 0.769]	0.000	-0.053
Zero-shot	Directional	Uncertainty Gate	0.127 [0.097, 0.158]	0.658 [0.521, 0.787]	0.000	-0.037
Zero-shot	Directional	Risk Filter	0.630 [0.577, 0.682]	0.690 [0.615, 0.764]	0.000	-0.260
Zero-shot	Directional	Cost-Aware Gate	0.175 [0.137, 0.215]	0.648 [0.515, 0.770]	0.000	-0.044
Zero-shot	Directional	Hard Role Gate	0.642 [0.590, 0.693]	0.694 [0.618, 0.767]	0.000	-0.262
Zero-shot	Directional	Lifecycle Checklist	0.218 [0.173, 0.265]	0.641 [0.517, 0.760]	0.000	-0.047
Zero-shot	Risk-limit	Direct Prior	0.272 [0.227, 0.318]	0.074 [0.021, 0.138]	0.294	-0.048
Zero-shot	Risk-limit	Confidence Gate	0.037 [0.020, 0.057]	0.045 [0.000, 0.158]	0.182	-0.005
Zero-shot	Risk-limit	Uncertainty Gate	0.045 [0.028, 0.065]	0.074 [0.000, 0.194]	0.185	-0.005
Zero-shot	Risk-limit	Risk Filter	0.263 [0.218, 0.312]	0.070 [0.014, 0.134]	0.291	-0.048
Zero-shot	Risk-limit	Cost-Aware Gate	0.022 [0.010, 0.035]	0.077 [0.000, 0.273]	0.231	-0.003
Zero-shot	Risk-limit	Hard Role Gate	0.192 [0.148, 0.238]	0.078 [0.016, 0.163]	0.000	-0.027
Zero-shot	Risk-limit	Lifecycle Checklist	0.023 [0.012, 0.038]	0.071 [0.000, 0.250]	0.000	-0.003
Recalibrated	Directional	Direct Prior	0.642 [0.590, 0.692]	0.694 [0.619, 0.765]	0.000	-0.262
Recalibrated	Directional	Confidence Gate	0.460 [0.410, 0.512]	0.678 [0.592, 0.760]	0.000	-0.187
Recalibrated	Directional	Uncertainty Gate	0.513 [0.462, 0.565]	0.688 [0.606, 0.769]	0.000	-0.203
Recalibrated	Directional	Risk Filter	0.560 [0.505, 0.615]	0.702 [0.623, 0.781]	0.000	-0.250
Recalibrated	Directional	Cost-Aware Gate	0.220 [0.177, 0.267]	0.644 [0.521, 0.764]	0.000	-0.047
Recalibrated	Directional	Hard Role Gate	0.642 [0.590, 0.692]	0.694 [0.619, 0.767]	0.000	-0.262
Recalibrated	Directional	Lifecycle Checklist	N/A	N/A	N/A	N/A
Recalibrated	Risk-limit	Direct Prior	0.272 [0.225, 0.318]	0.074 [0.020, 0.136]	0.294	-0.048
Recalibrated	Risk-limit	Confidence Gate	0.202 [0.162, 0.245]	0.050 [0.000, 0.115]	0.298	-0.025
Recalibrated	Risk-limit	Uncertainty Gate	0.227 [0.185, 0.270]	0.059 [0.009, 0.121]	0.301	-0.037
Recalibrated	Risk-limit	Risk Filter	0.247 [0.202, 0.293]	0.061 [0.012, 0.124]	0.297	-0.055
Recalibrated	Risk-limit	Cost-Aware Gate	0.037 [0.020, 0.057]	0.045 [0.000, 0.160]	0.318	-0.002
Recalibrated	Risk-limit	Hard Role Gate	0.192 [0.147, 0.237]	0.078 [0.017, 0.161]	0.000	-0.027
Recalibrated	Risk-limit	Lifecycle Checklist	N/A	N/A	N/A	N/A

Table 14: The prospective inverse-rank hypothesis is not supported. The primary row uses six shared directional rules. The registered leave-one-rule-out sensitivity shows perfect concordance after omitting Lifecycle; it is not a replacement primary endpoint. Exact p uses the preregistered inverse-or-more-extreme alternative.

Endpoint	Dates	Rules	ρ	Boot. ρ [95%]	τ_b	Exact p	$\Pr(\rho < 0)$	Valid boot.	Primary support
Primary: all rules	200	6	0.657	0.543 [-0.657, 0.943]	0.600	0.932	0.173	1.000	not supported
Sensitivity: omit Lifecycle	200	5	1.000	N/A	1.000	1.000	N/A	N/A	registered sensitivity

Table 15: Development-only threshold recalibration restores coverage but does not uniformly improve authority quality. Differences are recalibrated minus zero-shot on the one-time paper test. Rule forms remain unchanged. Lifecycle is structural N/A because none of 36 development candidates reached the 0.10 coverage floor; the floor was not relaxed.

Rule	Zero cov.	Recal. cov.	Δ cov.	Δ authority viol.	Δ utility	Status
Direct Prior	0.457	0.457	+0.000	+0.000	+0.000	unchanged
Confidence Gate	0.103	0.331	+0.228	+0.058	-0.077	calibrated
Uncertainty Gate	0.086	0.370	+0.284	+0.044	-0.099	calibrated
Risk Filter	0.447	0.403	-0.043	+0.005	+0.002	calibrated
Cost-Aware Gate	0.098	0.128	+0.030	+0.020	-0.001	calibrated
Hard Role Gate	0.417	0.417	+0.000	+0.000	+0.000	unchanged
Lifecycle Checklist	0.121	N/A	N/A	N/A	N/A	NO_ELIGIBLE_CANDIDATE_COVERAGE_FLOOR

Table 16: Source-format quality is reported separately from authorization quality. Rows describe proposal-source behavior under the same task, not a model leaderboard. Review flag is model-recommended metadata and is not direct execution evidence.

Proposal source	Task	Dates	Raw valid	Malformed	Repairs	Abstain	Mean conf.	Review flag	Harm overconf.
claude-sonnet-4-6	Directional	200	1.000	0.000	0	0.460	0.591	0.270	0.286
gpt-5.4-mini	Directional	200	0.995	0.005	0	0.215	0.682	0.305	0.274
gpt-5.5	Directional	200	1.000	0.000	0	0.400	0.652	0.265	0.242
claude-sonnet-4-6	Risk-limit	200	1.000	0.000	0	0.815	0.654	0.205	0.000
gpt-5.4-mini	Risk-limit	200	0.995	0.005	0	0.550	0.684	0.325	0.067
gpt-5.5	Risk-limit	200	1.000	0.000	0	0.820	0.678	0.575	0.000

G TRAINING UTILITY DETAILS

Each D0–D7 variant uses matched cluster and hyperparameter budgets. The primary learner is logistic regression with $C \in \{0.1, 1, 10\}$ selected on seen validation mechanisms. The threshold is also selected on validation only. The four endpoint mechanisms are severe controlled stress, stress-only-period controlled rows, provenance

Table 17: Mechanism coverage invalidates most training endpoints (validation only). D0 is outside the data-utility Holm family; the role-neutral control is a negative control.

Variant	Valid endpoints	Interpretation
D0	5/5	prediction diagnostic
D1	0/5	authorization reference
D2	0/5	authorization variant
D3	0/5	authorization variant
D4	0/5	authorization variant
D5	1/5	authorization variant
D6	0/5	authorization variant
D7	0/5	authorization variant
D7 role-neutral	0/5	negative control; outside Holm

Table 18: Holm-adjusted D2–D7 versus D1 validation contrasts. Delta FAR is N/A because no candidate/reference pair has valid endpoints in the same seed.

Contrast	Delta FAR	Valid pairs	Raw p	Holm p	Reject
D2 vs D1	N/A	0	1.000	1.000	no
D3 vs D1	N/A	0	1.000	1.000	no
D4 vs D1	N/A	0	1.000	1.000	no
D5 vs D1	N/A	0	1.000	1.000	no
D6 vs D1	N/A	0	1.000	1.000	no
D7 vs D1	N/A	0	1.000	1.000	no

paraphrase, and provenance multi-hop. A seed endpoint is valid only when all four coverage LCBs are at least 0.05.

The D7 role-neutral control masks role and lineage fields during training, calibration, and evaluation. It has 0/5 valid endpoints. Feature-group masking also shows that the fixed learner depends materially on role features: masking them raises D7 validation-unseen FAR to 0.167 and ALR to 0.165. These are shortcut diagnostics, not causal feature attributions.

The prospective v0.3 robustness study uses Logistic Regression, histogram gradient boosting, MLP, and selective Logistic; budgets of 400, 1,000, 5,000, and 20,000 independent clusters; controlled-seen, multi-generator, and full-audit-seen curricula; and three frozen seeds. At 20,000 clusters, full-audit Logistic and selective Logistic reach mean valid-mechanism fraction 0.833, but neither has all six mechanisms valid in any seed. Sequential-market validation remains below the fixed 0.05 coverage floor, so complete primary FAR remains N/A. This distinguishes partial endpoint recovery from a valid cross-mechanism comparison.

H PAPER-TEST STABILITY AND GENERATOR SENSITIVITY

The paper split reproduces validation rankings, while the prospective mechanisms test whether that stability extends beyond one generator family.

Sequential and institutional generators have raw FAR-rank Spearman 0.994; their correlations with utility IID are 0.683 and 0.738.

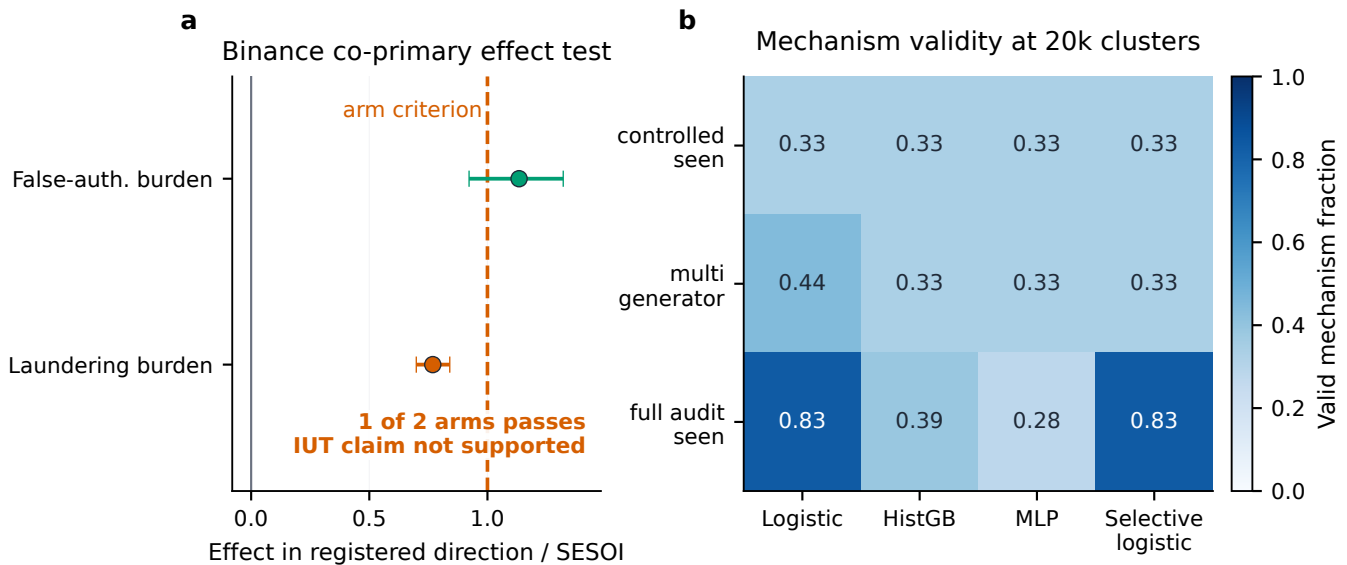


Figure 5: Validity gates preserve partial and undefined evidence. Panel (a) divides registered Binance effects by their SESOIs; error bars are 95% cluster-bootstrap intervals and the dashed line is the arm criterion. Planning power uses frozen v0.2 variance. Only one arm passes. Panel (b) reports the fraction of six mechanisms with coverage LCB at least 0.05 at 20,000 clusters; no cell reaches one.

Table 19: Validation-to-paper and cross-generator rank stability. Constant certification ranks remain N/A and are not shown.

Comparison	Rank	Spearman
Validation vs paper test	raw FAR	1.000
Validation vs paper test	certification	1.000
Institutional vs Sequential	raw FAR	0.994
Institutional vs Utility IID	raw FAR	0.738
Sequential vs Utility IID	raw FAR	0.683

Lowest-FAR and highest-certification rules differ in two of three families. The controlled generators cannot establish a universal financial relationship; they supply interpretable, mechanistically distinct scenarios for auditing rule-by-generator interactions.

I REVIEW WORKLOAD AND BASELINE GOVERNANCE

Review routing is not a free abstention channel. On validation, Lifecycle Checklist sends 43.5% of controlled rows to review and exceeds a 40% review-capacity scenario by 1,383 rows. Of those reviewed cases, 79.7% would be harmful under direct execution, but only 1.9% contain a safe positive opportunity. The paper-test review rate is 42.9%. These are simulated workloads and normalized cost sensitivities, not measured staffing time or reviewer accuracy.

The baseline-governance audit reproduces the registered Lifecycle rule exactly, then evaluates frozen component removals and independently motivated compositions. Role + Confidence + Cost Gate and Lifecycle without the reduce route both achieve validation

certification volume 0.60, above the registered Lifecycle value 0.30. Continuous conservative hypervolume instead ranks Hard Role Gate above Lifecycle. These results reject a Lifecycle-only winner narrative and support frontier reporting.

J BASELINE FIELD ACCESS

Deployable rules may use candidate action, confidence, uncertainty, expected edge, cost and liquidity proxies, horizon, current source role, and legal lineage fields declared for their task. They may not use realized returns, post-cost utility, harm labels, oracle opportunity, laundering truth, tail loss, future decoys, or test outcomes. Oracle-positive opportunity and attack labels are evaluation-only. Public rationales, profiles, and user identities are not execution evidence.

The equal-information EPV Adapter uses the same legal fields as other baselines. It is included to test whether low FAR transfers to provenance safety, not to establish or refute the companion method paper. The overlap audit snapshots the current AAAI source and checks seeds, exact data hashes, exact/perceptual figure hashes, headline claims, and shared 8/12-grams. The current audit reports zero conflicts across those gates; a fresh snapshot is required before submission.

K NEGATIVE RESULTS AND CHANGE POLICY

FinAuth-Audit keeps development failures as audit evidence. The first controlled generator allowed non-opportunity noise to create unintended positive opportunities; a rule-independent correction repaired the density control. The first training endpoint averaged mechanisms before checking per-mechanism coverage and produced a misleading low-FAR result; the registered endpoint now

requires every mechanism to pass its coverage floor. Strict verification later found a cross-layer holdout leak in D4/D5/D7, after which both controlled and provenance holdout families were applied and the affected corpora rebuilt. No test outcome was opened during these changes.

After validation development, the existing test clusters were partitioned using event IDs only. A validation rehearsal reproduced registered outputs, the scientific surface was frozen in a content-addressed archive, and the paper-test evaluator was executed once. After inspection, generator distributions, attack definitions, seeds, feature manifests, certification thresholds, primary endpoints, and robustness profiles cannot change to obtain a preferred ranking. A correction requires a reproducible defect and a versioned record that

preserves the original result. The 5,000-cluster community-hidden set remains unevaluated.

L ETHICS AND RESPONSIBLE USE

Controlled rows are synthetic authorization scenarios and can encode modeling assumptions or omit institutional constraints. Public rows contain event-market observations, not recommendations. The artifact must not be used to justify live trading, automated approval, or removal of human oversight. Users should report source provenance, decision-time legality, coverage, review load, and structural N/A; publishing FAR alone is insufficient. Any institutional extension should add expert review, privacy controls, jurisdiction-specific policy, and post-launch monitoring before operational use.